
PROFESSIONAL PORTFOLIO · 2026

MUHAMMAD FATHONI

Electrical Engineering Graduate

IT · AI · Electrical Engineer · IoT · Data Science

Universitas Islam Indonesia · GPA 3.78 / 4.00



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01 · ABOUT ME

About Me

I am a final-year Electrical Engineering student at Universitas Islam Indonesia with a deep enthusiasm for the intersection of power systems, embedded technology, and modern artificial intelligence. As a full scholarship awardee through the UII Excellent Community program, I have consistently maintained academic excellence while actively engaging in research, leadership, and entrepreneurial initiatives.

My professional interests span Electrical Engineering, Internet of Things (IoT), Machine Learning, and Generative AI applications in industry. I have hands-on experience designing end-to-end smart systems, combining hardware (ESP32, sensors, PLC, microcontrollers) with predictive ML models and large language model integrations. Beyond technical work, I bring proven leadership in student organizations and a track record of competition wins and research grants.

I am driven, adaptive, and committed to continuous learning. I thrive in collaborative, forward-thinking environments where I can contribute to meaningful innovation while growing alongside experienced professionals.



CAREER OBJECTIVE

To contribute to a forward-thinking organization as an Engineer, Data, or AI/ML specialist — leveraging my electrical engineering foundation, programming skills, and passion for innovation to build impactful, intelligent industrial solutions.

02 · EDUCATION

Education & Academic Records

Universitas Islam Indonesia (UII)

Aug 2022 – Present

Bachelor of Electrical Engineering · Faculty of Industrial Technology

GPA 3.78 / 4.00

140 SKS Completed · UII Excellent Community Scholarship

KEY COURSEWORK

- Power Systems
- Control Systems
- PLC & SCADA
- Renewable Energy
- Power Electronics
- Embedded Systems
- Machine Learning
- Digital Systems (VHDL/FPGA)
- Microprocessor Systems
- Telecommunications
- Computer Networks
- Engineering Mathematics

03 · SKILLS

Technical Skills

HARD SKILLS

POWER & ELECTRICAL SYSTEMS

- Electrical Troubleshooting & Wiring
- Circuit Analysis (Analog & Digital)
- Power System Reliability (SAIDI/SAIFI)
- PLC Programming · PID Control · LabVIEW

EMBEDDED & IOT

- Microcontrollers (ESP32, Arduino, ATmega)
- Sensors (PZEM-004T, DHT, MQ Series)
- MQTT Protocol & Real-Time Telemetry
- Proteus, CodeVision AVR, VHDL/FPGA

PROGRAMMING & DATA

- Python, C, C++, HTML, CSS, JS
- Machine Learning (Scikit-learn, SVM, Random Forest)
- Generative AI (Google Gemini API)
- Pandas, Matplotlib, Firebase

TOOLS & NETWORKING

- Microsoft Office Suite
- Computer Networking (HCIA-Datacom)
- Project Management · Git

SOFT SKILLS

Critical & Analytical Thinking

Problem Solving

Teamwork & Collaboration

Communication & Public Speaking

Leadership & Coordination

Time Management

Adaptability & Fast Learning

Detail-Oriented & Initiative

LANGUAGES

Bahasa Indonesia — *Native*
English — *Professional Working Proficiency*

04 · PROJECTS

FINAL PROJECT · 2025

SEMS — Smart Energy Management System

Sole Developer (Hardware, Firmware, ML, Web, AI Integration)



ESP32

IoT

Random Forest

Generative AI

Firebase

An end-to-end smart energy monitoring and recommendation platform that combines real-time power telemetry, predictive machine learning, and generative AI to help Indonesian households reduce electricity consumption. The system integrates ESP32-based hardware, a Random Forest forecasting model, Firebase real-time database, and Google Gemini 2.5 Flash to deliver context-aware energy advice.

KEY HIGHLIGHTS

- Real-time power monitoring with ESP32 + PZEM-004T + DHT22 sensors over MQTT
- Random Forest model achieved R^2 0.9893 (98.93% accuracy) for monthly forecasting
- Google Gemini 2.5 Flash integration for personalized energy-saving recommendations
- Firebase dashboard with PLN tariff calculator (450VA – 5500VA customer profiles)

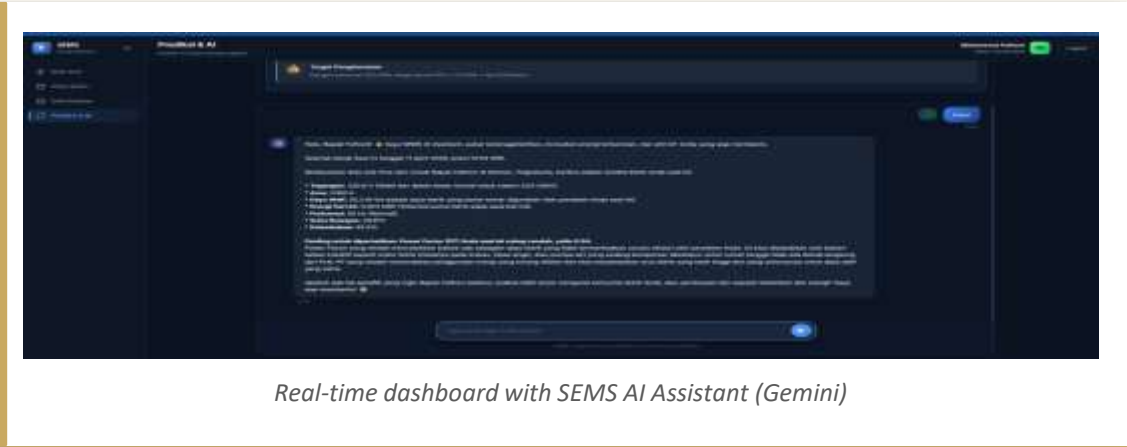
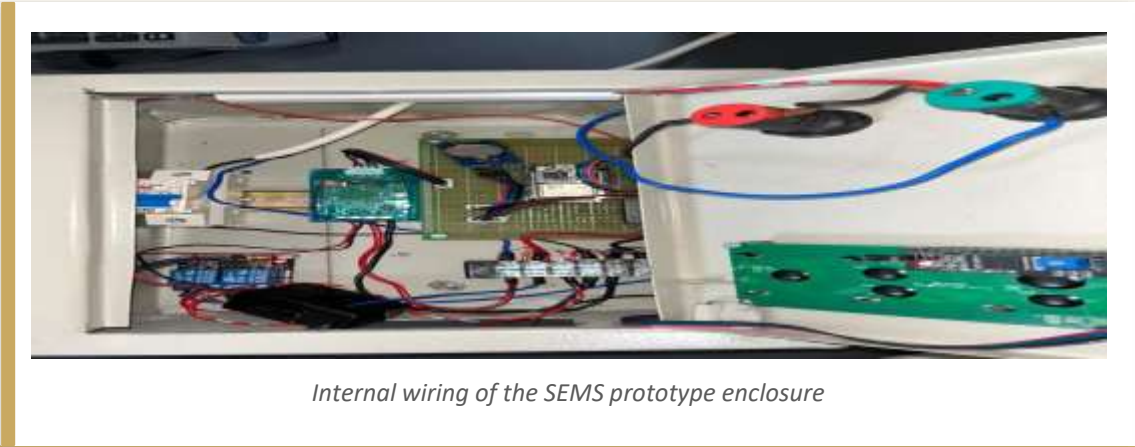
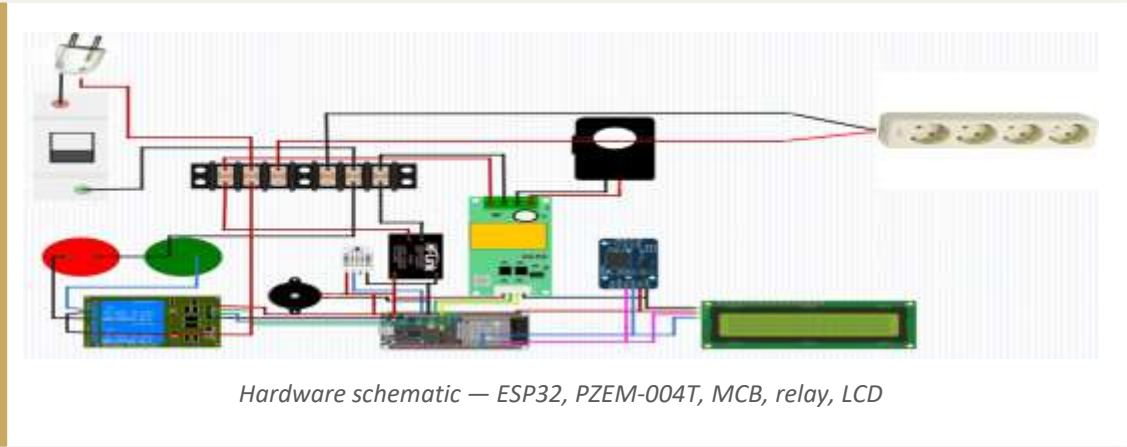
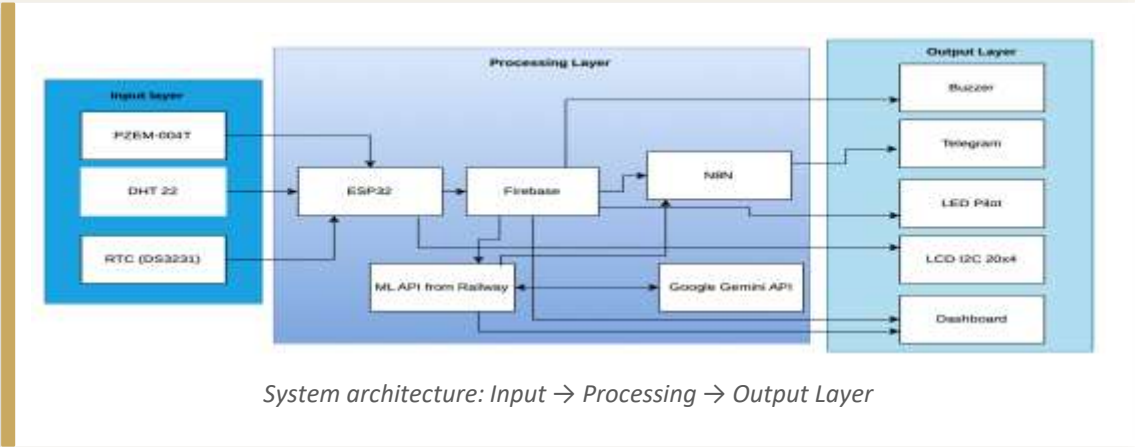


SEMS device — front panel with LCD display

04 · PROJECTS (CONT.)

SEMS — System Architecture & Hardware

A complete IoT pipeline: from physical sensors and microcontroller to cloud database, ML inference, and AI-powered dashboard.



04 · PROJECTS (CONT.)

PERSONAL PRODUCT · NOORA DEEN TECHNOLOGY · 2026

Noora: Muslim Habit-Building App

Co-Founder & Sole Engineer (Mobile, Backend, UI)

Capacitor 8

JavaScript

Supabase

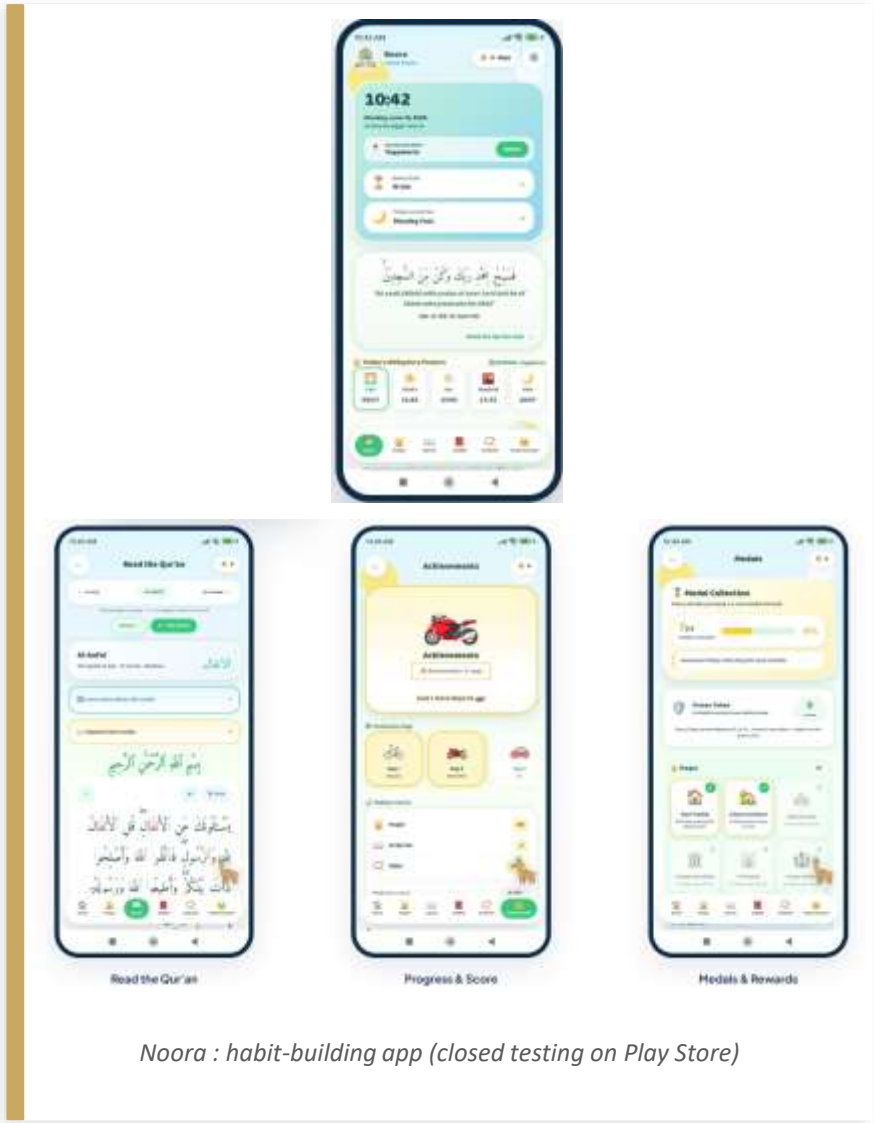
REST API

Android

A mobile application that helps Muslims build consistent daily worship habits in a positive, engaging way. Noora combines an international prayer-time engine, a complete digital Qur'an with an authentic 604-page Madinah mushaf mode, and a gamified “Garden” that grows with the user's consistency. Built as the flagship product of Noora Deen Technology, it runs fully online and offline, and is currently in closed testing on the Google Play Store.

KEY HIGHLIGHTS

- Region-aware prayer times: Kemenag for Indonesia, auto-switching to JAKIM, MUIS, Umm al-Qura, ISNA or Diyanet abroad
- Full Qur'an with verified Saheeh International translation, tafsir, and a 604-page Madinah mushaf with page-turn animation
- Gamified habit engine: per-category and combined streaks, freeze tokens, an evolving garden and claimable achievements
- Works fully offline, or syncs progress across devices in real time via Supabase; custom real-time ID/EN i18n engine



Noora : habit-building app (closed testing on Play Store)

04 · PROJECTS (CONT.)

CORPORATE WEBSITE · NOORA DEEN TECHNOLOGY · 2026

Noora Deen Technology: Company Website

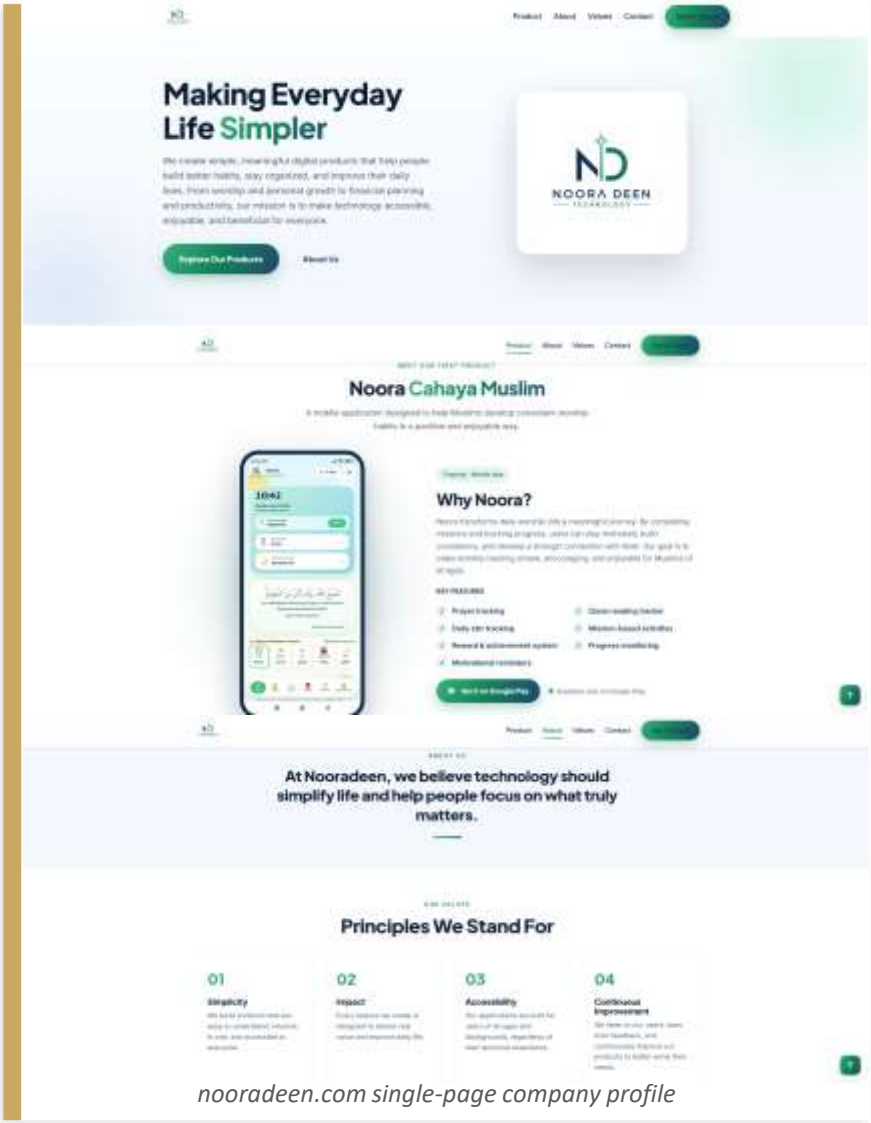
Designer & Developer (Front-End, Branding)

- HTML
- CSS
- JavaScript
- Responsive
- Accessibility

A lightweight, fast single-page company profile for Noora Deen Technology, built entirely with vanilla HTML, CSS and JavaScript, no framework, no build step. The site presents the company's mission (“Making Everyday Life Simpler”), its products, values and contact channels, and is engineered for accessibility and smooth performance on any device.

KEY HIGHLIGHTS

- Fully responsive with a sticky navbar, hamburger menu and scroll-spy navigation across desktop, tablet and mobile
- Scroll-reveal animations and animated statistic counters built without any external animation library
- Accessibility-first: keyboard navigation, ARIA labels, and respect for prefers-reduced-motion
- Custom brand system: navy-and-green palette, north-star logo with an “ND” monogram, zero-backend contact form



04 · PROJECTS (CONT.)

PERSONAL PRODUCT · AI FINANCE · 2026

Saku: AI-Powered Personal Finance Manager

Sole Developer (Front-End, Backend, AI, Mobile)

React 19

TypeScript

Gemini 2.5

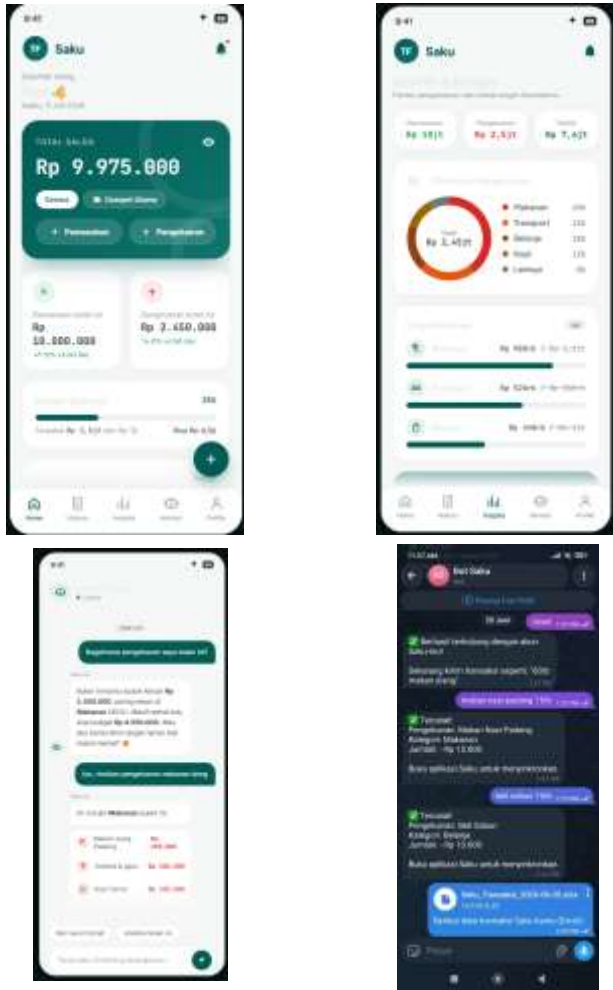
Express

Capacitor

A mobile-first personal finance manager for Indonesian users. Saku makes logging income and expenses nearly frictionless, then layers on an AI advisor (Google Gemini 2.5 Flash) that is grounded on the user's own transactions and replies in natural Bahasa Indonesia. A Telegram bot lets users record spending simply by chatting, “60rb makan siang” becomes a categorised expense of Rp 60.000.

KEY HIGHLIGHTS

- Telegram bot logs spending from natural language: “60rb makan siang” becomes a categorised Rp 60.000 expense
- Grounded AI advisor: WhatsApp-style chat answering questions about the user's real transactions in Bahasa Indonesia
- Analytics suite: category donut, 7-day income-vs-expense trends, budget progress and AI-generated insights
- Cross-platform React 19 + Vite client wrapped with Capacitor 8, deployed serverless on Vercel with Excel export



Saku dashboard, analytics and AI advisor

04 · PROJECTS (CONT.)

MACHINE LEARNING COURSE · 2024

Iris Flower Classification with Support Vector Machine

Developer



Python

SVM

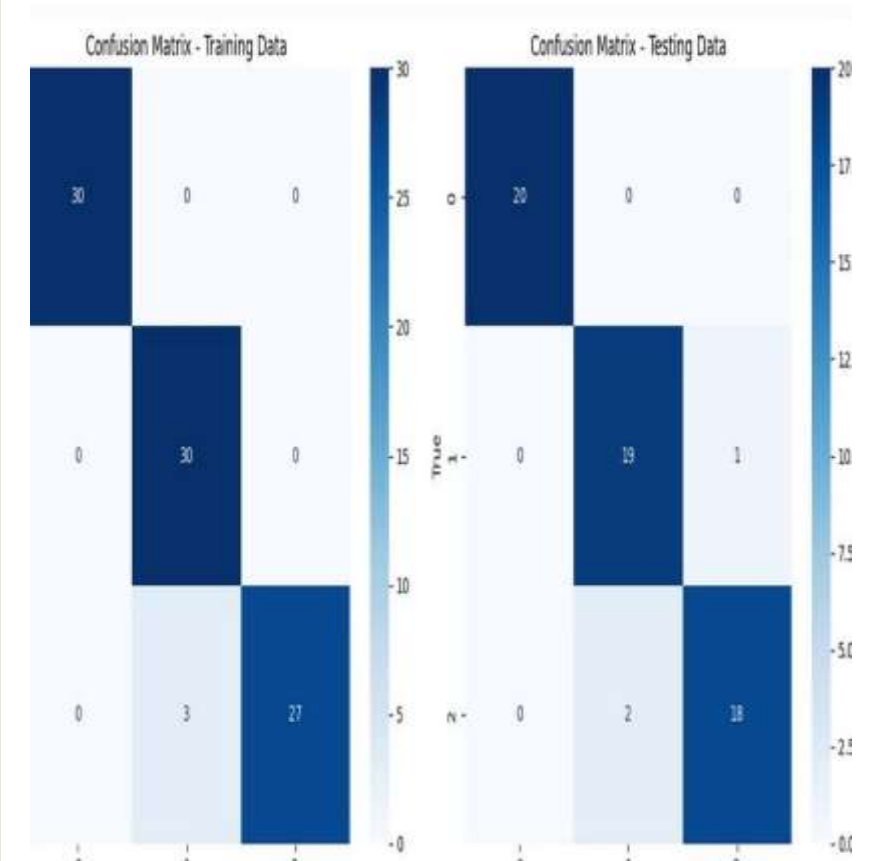
Scikit-learn

Google Colab

Built and evaluated a Support Vector Machine model to classify Iris flower species using petal and sepal measurements. The project followed a complete ML pipeline — from exploratory data analysis and preprocessing through model training, evaluation with confusion matrices, and classification reporting — strengthening my foundation in supervised learning workflows.

KEY HIGHLIGHTS

- Achieved over 90% classification accuracy on test set with linear-kernel SVM
- Train-test split of 80:20 with proper cross-validation methodology
- Full ML pipeline: preprocessing, training, evaluation, and reporting
- Practical foundation for advanced supervised learning workflows



Confusion matrix — training vs testing performance

04 · PROJECTS (CONT.)

PKM-KC FUNDED PROJECT · 2024

AromatheraWrist — Smart Bracelet for Anxiety Relief

Lead Developer · PKM-KC Team

Wearable

Biomedical

Mobile App

Embedded

An innovative wearable bracelet that integrates aromatherapy delivery with a real-time pulse sensor, controlled via a companion mobile application to help reduce anxiety symptoms in users. The project combined biomedical signal processing with user-centered design, and was awarded PKM Karsa Cipta funding by the Indonesian Ministry of Education.

KEY HIGHLIGHTS

- Awarded PKM Karsa Cipta funding by Indonesian Ministry of Education (2024)
- 4th Place — PKM IC Karsa Cipta Scheme
- Real-time pulse rate monitoring with embedded sensor system
- Mobile app integration for aromatherapy spray control



AromatheraWrist prototype — aromatherapy spray and pulse sensor

04 · PROJECTS (CONT.)

ELECTRONICS SYSTEMS PROJECT · 2024

IoT Automatic Temperature Control



Developer

ESP32

MQTT

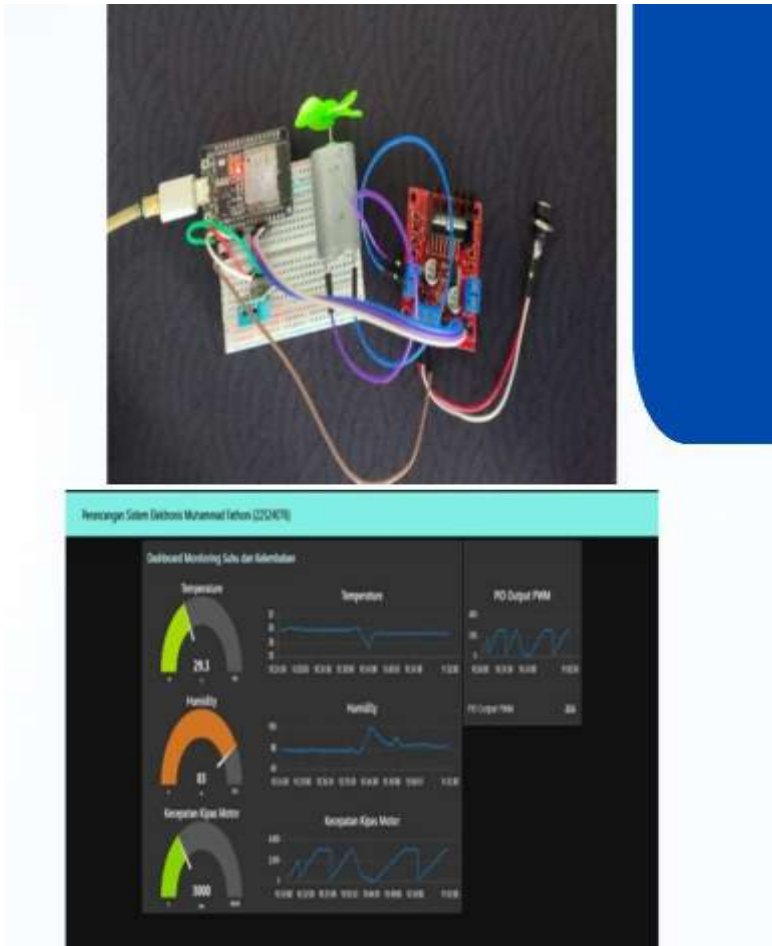
PID Control

Python

An IoT-based automatic temperature control system using ESP32 and DHT11 sensor, with a Python-based PID controller communicating via MQTT to dynamically regulate fan speed based on real-time room temperature. The system demonstrated stable closed-loop control with accurate response to environmental changes.

KEY HIGHLIGHTS

- ESP32 (publisher) + Python application (subscriber) over MQTT broker
- L298N motor driver controlling DC fan with PID-tuned response
- Real-time monitoring dashboard with temperature, humidity, and fan speed gauges
- Demonstrated stability and accuracy in real-time control loop



Hardware setup + monitoring dashboard with PID output

04 · PROJECTS (CONT.)

EMBEDDED SYSTEM FINAL PROJECT · 2024

Air Quality Monitoring Tool

Developer

Arduino Uno

NodeMCU ESP8266

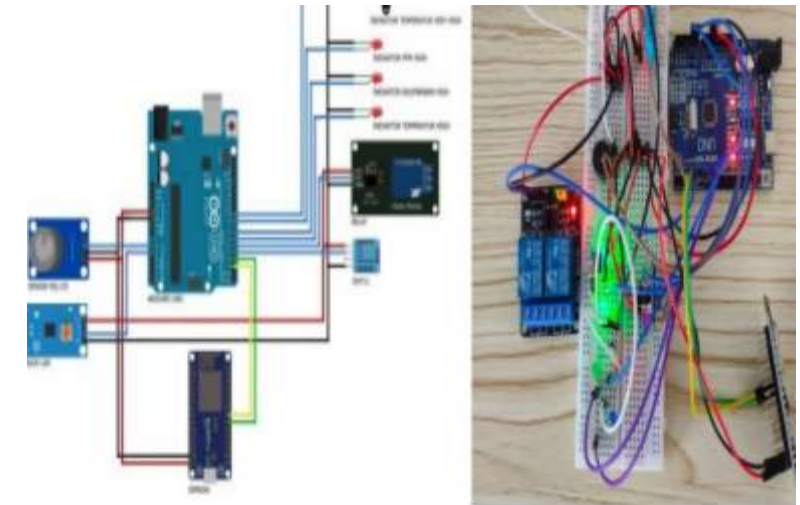
Blynk IoT

Multi-Sensor

A real-time air quality monitoring system that measures temperature, humidity, gas concentration (PPM), and light intensity using multiple sensors integrated with Arduino Uno and NodeMCU ESP8266. Data is transmitted to the Blynk IoT platform, enabling real-time monitoring and remote control via smartphone application.

KEY HIGHLIGHTS

- Multi-sensor integration: DHT11 (temp/humidity), MQ-135 (gas), LDR (light)
- Cloud-based dashboard with mobile remote control via Blynk IoT
- Real-time environmental data logging with notification alerts
- Day/night mode detection with automatic relay switching



Hardware schematic, breadboard prototype, and Blynk app

04 · PROJECTS (CONT.)

MICROPROCESSOR SYSTEMS PROJECT · 2024

Three-Lane Adaptive Traffic Light Simulator



Developer

ATmega328P

CodeVision AVR

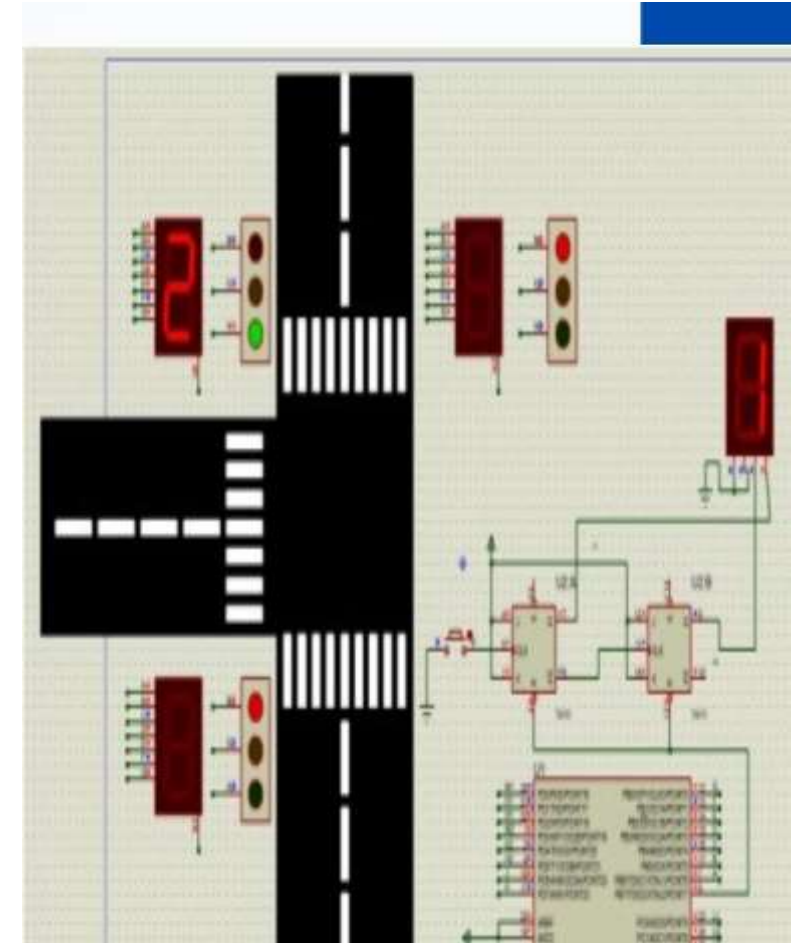
Proteus

C

An adaptive traffic light controller for a three-way intersection, programmed in C via CodeVision AVR (CVAVR) and simulated using Proteus 8 Professional. Each lane is equipped with a seven-segment countdown timer, and the system dynamically adjusts the green light duration based on traffic density signals from push buttons combined with JK flip-flop logic.

KEY HIGHLIGHTS

- ATmega328P microcontroller programmed in C via CVAVR
- 7-segment countdown display per lane with dynamic timing
- Density-aware green-phase scheduling using JK flip-flop logic
- Simulated and validated in Proteus 8 Professional



Three-way intersection simulation in Proteus 8

05 · EXPERIENCE

Professional Experience

Engineering Intern

PT PLN (Persero) — ULP Pasir Pengaraian

- Worked in the Engineering (Teknik) Division, handling customer outage data through the APKT system.
- Performed reliability analysis using SAIDI and SAIFI indices to evaluate distribution network performance.
- Supported division performance evaluation and collaborated with field technicians on outage resolution.
- Gained hands-on understanding of PLN distribution operations, work culture, and customer service standards.

Sep 2025 – Oct 2025

Riau, Indonesia



Laboratory Assistant — Telecommunication Systems

Department of Electrical Engineering, Universitas Islam Indonesia

- Guide undergraduate students through hands-on telecommunication systems experiments.
- Prepare lab modules, evaluate student reports, and assist with troubleshooting.

Apr 2025 – Present

Yogyakarta



Laboratory Assistant — Control Systems (PID & LabVIEW)

Department of Electrical Engineering, Universitas Islam Indonesia

- Mentored students on PID controller design, tuning, and LabVIEW-based simulation.
- Assisted in conducting practical sessions for closed-loop control experiments.

2025

Yogyakarta



05 · EXPERIENCE (CONT.)

Professional Experience

Laboratory Assistant — Digital Systems (VHDL & FPGA)

Department of Electrical Engineering, Universitas Islam Indonesia

- Mentored students on digital logic design using VHDL and FPGA implementation.
- Supervised hands-on laboratory work in combinational and sequential logic circuits.

Laboratory Assistant — Electric Circuit II

Department of Electrical Engineering, Universitas Islam Indonesia

- Mentored second-year students on AC circuit analysis, network theorems, and resonance.
- Conducted lab assessments and supported instructors during practical sessions.



Lab session — undergraduate practical

TEACHING PROFICIENCY

Across four semesters as a laboratory assistant, I have mentored over 200 undergraduate students in core electrical engineering subjects — from fundamental circuit theory to modern control systems and digital design. This experience has sharpened my technical communication skills, patience, and ability to break down complex concepts.

06 · LEADERSHIP

Leadership & Organizations

IEEE STUDENT BRANCH UII

- Director of Academies
Jun 2024 – Present
- Academies Staff
Mar 2023 – Jun 2024
- Member (IEEE Global)
Jan 2024 – Dec 2024

UASC ELECTRIC VEHICLE

- Head of Programming Division
May 2024 – Mar 2025
- Electrical Engineer Staff
Dec 2023 – May 2024
- Electrical Engineer Intern
Jul 2023 – Dec 2023

UNISI ROBOTIC UII

- Electrical Division Staff
May 2024 – Present

UII EXCELLENT COMMUNITY (UII EC)

- Full Scholarship Awardee
Aug 2022 – Present

COMMITTEE EXPERIENCE

Electrical Link Togetherness (ELITE) — Master of Ceremony
Events Department · Department of Electrical Engineering UII
Hosted department-wide events; coordinated event flow and on-stage communication.

07 · ACHIEVEMENTS AND CERTIFICATIONS

Achievements

- 1ST

1st Place — IT Centrum Startup Hackathon 2024

Universitas Islam Indonesia · Dec 13–14, 2024
- 1ST

1st Place — Business Idea Competition

Student Entrepreneur Day 2024
- 4TH

4th Place — PKM IC Karsa Cipta Scheme

Ministry of Education · 2024
- ★

PKM Karsa Cipta Funding Recipient

Ministry of Education · 2024
- 3RD

3rd Place — PKM IC Community Service Scheme

Ministry of Education · 2025
- 5TH

5th Place — PKM IC Innovative Work Scheme

Ministry of Education · 2025
- ★

UII Excellent Community Full Scholarship Awardee

UII · 2022 – 2026

Certifications

HCIA-Datacom V1.0 Huawei Talent Online	Dec 13, 2023
Basic PLC Training Dept. of Electrical Eng., UII	Dec 21, 2024
Python for Data Science (Beginner) Dept. of Electrical Eng., UII	Nov 2024 & May 2025
Leadership Training UII EC 2022 UII Student Affairs Directorate	Jun 29, 2024
Personal Development Basic Training UII Student Affairs Directorate	Jun 25, 2023
IEEE Leadership Class 2022 IEEE Student Branch UII	Nov–Dec 2022

08 · CONTACT

Let's Work Together.

I'm open to engineering, AI/ML, and data roles, and ready to contribute.

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LOCATION

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THANK YOU FOR YOUR CONSIDERATION